

```
1 export const SOFTWARE_NAME = '天平实验训练示例';
2 export const VERSION = 'V1.0';
3 export const OWNER = '示例著作权人';
4 export const STEP_NAMES = [
5   '检查实验器材',
6   '游码归零',
7   '空载调平',
8   '左盘放置物体',
9   '由大到小添加砝码',
10  '游码微调平衡',
11  '读取并记录质量',
12  '整理器材归位',
13 ];
14 export const STEP_POINTS = [10, 10, 15, 15, 20, 10, 10, 10];
15 export const WEIGHTS = [500, 200, 100, 50]; // 0.1 g units
16 export interface Step {
17   name: string;
18   max: number;
19   passed: boolean;
20   score: number;
21 }
22 export interface Event {
23   at: string;
24   action: string;
25   message: string;
26   violation: boolean;
27 }
28 export interface Session {
29   id: string;
30   label: string;
31   source: 'practice' | 'demo';
32   createdAt: string;
33   updatedAt: string;
34   finishedAt: string | null;
35   status: 'active' | 'finished';
36   checked: boolean;
37   calibrated: boolean;
38   rider: number;
39   nut: number;
40   objectSide: 'box' | 'left' | 'right';
41   objectMass: number;
42   weights: { mass: number; side: 'left' | 'right' }[];
43   tool: 'tweezers' | 'hand';
44   steps: Step[];
45   events: Event[];
46   reading: number | null;
47   score: number;
48   feedback: string;
49   balanced: boolean;
50   measurement?: { weightTotal: number; rider: number; value: number; at: string };
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51   difference: number;
52 }
53 export type Action =
54   | { type: 'check' }
55   | { type: 'rider'; value: number }
56   | { type: 'nut'; value: number }
57   | { type: 'tool'; value: 'tweezers' | 'hand' }
58   | { type: 'object'; side: 'box' | 'left' | 'right' }
59   | { type: 'addWeight'; mass: number; side: 'left' | 'right' }
60   | { type: 'removeWeight'; index: number }
61   | { type: 'read'; value: number }
62   | { type: 'tidy' };
63 export interface ApiError {
64   error: string;
65 }
66 import { createServer as httpServer } from 'node:http';
67 import type { IncomingMessage, ServerResponse } from 'node:http';
68 import { readFile } from 'node:fs/promises';
69 import { resolve, extname, sep } from 'node:path';
70 import { SOFTWARE_NAME, VERSION } from '../shared/types.ts';
71 import {
72   applyAction,
73   createSession,
74   finishSession,
75   InputError,
76   validateAction,
77 } from './engine.ts';
78 import { Store } from './storage.ts';
79 class HttpError extends Error {
80   status: number;
81   constructor(status: number, message: string) {
82     super(message);
83     this.status = status;
84   }
85 }
86 function json(res: ServerResponse, status: number, value: unknown) {
87   res.writeHead(status, {
88     'Content-Type': 'application/json; charset=utf-8',
89     'Cache-Control': 'no-store',
90   });
91   res.end(JSON.stringify(value));
92 }
93 async function body(req: IncomingMessage): Promise<unknown> {
94   if (!(req.headers['content-type'] || '').startsWith('application/json'))
95     throw new HttpError(415, '请使用 application/json 请求。');
96   let length = 0;
97   const chunks: Buffer[] = [];
98   for await (const chunk of req) {
99     length += chunk.length;
100    if (length > 16384) throw new HttpError(413, '请求体过大。');
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101     chunks.push(Buffer.from(chunk));
102   }
103   try {
104     return JSON.parse(Buffer.concat(chunks).toString('utf8'));
105   } catch {
106     throw new HttpError(400, '请求 JSON 格式错误。');
107   }
108 }
109 export function createServer(dataDir: string, distDir: string) {
110   const store = new Store(dataDir);
111   const root = resolve(distDir);
112   return httpServer(async (req, res) => {
113     res.setHeader('X-Content-Type-Options', 'nosniff');
114     res.setHeader('X-Frame-Options', 'DENY');
115     res.setHeader('Referrer-Policy', 'no-referrer');
116     try {
117       const address = req.socket.localPort;
118       const allowed = [`127.0.0.1:${address}`, `localhost:${address}`];
119       if (!allowed.includes(req.headers.host || ''))
120         throw new HttpError(403, '仅允许本机访问。');
121       if (req.method !== 'GET' && req.method !== 'HEAD') {
122         if (
123           req.headers.origin &&
124           !allowed.map((host) => `http://${host}`).includes(req.headers.origin)
125         )
126           throw new HttpError(403, '不允许其他网站修改本机实验记录。');
127         if (req.headers['sec-fetch-site'] === 'cross-site')
128           throw new HttpError(403, '拒绝跨站请求。');
129       }
130       const url = new URL(req.url || '/', `http://${req.headers.host}`);
131       const path = url.pathname;
132       if (path === '/api/health' && req.method === 'GET') {
133         await store.list();
134         return json(res, 200, { name: SOFTWARE_NAME, version: VERSION, status: 'ok' });
135       }
136       if (path === '/api/sessions' && req.method === 'GET')
137         return json(res, 200, (await store.list()).reverse());
138       if (path === '/api/sessions' && req.method === 'POST') {
139         const input = (await body(req)) as { label?: unknown; source?: unknown };
140         if (
141           !input ||
142           typeof input !== 'object' ||
143           (input.label !== undefined &&
144             (typeof input.label !== 'string' || input.label.trim().length > 80)) ||
145           (input.source !== undefined &&
146             (typeof input.source !== 'string' || ![ 'practice', 'demo' ].includes(input.source)))
147         )
148           throw new HttpError(400, '实验名称最多 80 字，来源须为 practice 或 demo。');
149         const created = await store.mutate((sessions) => {
150           const s = createSession(
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151         typeof input.label === 'string' && input.label.trim()
152         ? input.label.trim()
153         : undefined,
154         input.source as 'practice' | 'demo' | undefined,
155     );
156     sessions.push(s);
157     return s;
158 });
159 return json(res, 201, created);
160 }
161 const match = path.match(
162     /^\/api\/sessions\/([\w-]+)(?:\/(actions|finish|export))?$/,
163 );
164 if (match) {
165     const [, id, operation] = match;
166     if (req.method === 'GET' && (!operation || operation === 'export')) {
167         const session = (await store.list()).find((s) => s.id === id);
168         if (!session) throw new HttpError(404, '实验记录不存在。');
169         if (operation === 'export')
170             res.setHeader(
171                 'Content-Disposition',
172                 `attachment; filename="experiment-${id}.json"`,
173             );
174         return json(res, 200, session);
175     }
176     if (req.method === 'POST' && ['actions', 'finish'].includes(operation)) {
177         const input = await body(req);
178         if (operation === 'actions') validateAction(input);
179         const session = await store.mutate((sessions) => {
180             const index = sessions.findIndex((s) => s.id === id);
181             if (index === -1) throw new HttpError(404, '实验记录不存在。');
182             if (operation === 'actions') {
183                 validateAction(input);
184                 sessions[index] = applyAction(sessions[index], input);
185             } else sessions[index] = finishSession(sessions[index]);
186             return sessions[index];
187         });
188         return json(res, 200, session);
189     }
190     throw new HttpError(405, '不支持此请求方法。');
191 }
192 if (path.startsWith('/api/')) throw new HttpError(404, '接口不存在。');
193 if (!['GET', 'HEAD'].includes(req.method || ''))
194     throw new HttpError(405, '不支持此请求方法。');
195 let decoded: string;
196 try {
197     decoded = decodeURIComponent(path);
198 } catch {
199     throw new HttpError(400, '路径编码无效。');
200 }
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201     if (decoded.includes('\0') || decoded.includes('\'))
202         throw new HttpError(400, '路径无效。');
203     let file = resolve(root, `.${decoded}`);
204     if (file !== root && !file.startsWith(root + sep))
205         throw new HttpError(403, '禁止访问此路径。');
206     if (decoded === '/' || !extname(decoded)) file = resolve(root, 'index.html');
207     let content: Buffer;
208     try {
209         content = await readFile(file);
210     } catch {
211         throw new HttpError(404, '页面不存在, 请先运行构建脚本。');
212     }
213     const mime: Record<string, string> = {
214         '.html': 'text/html; charset=utf-8',
215         '.js': 'text/javascript; charset=utf-8',
216         '.css': 'text/css; charset=utf-8',
217         '.svg': 'image/svg+xml',
218         '.png': 'image/png',
219         '.ico': 'image/x-icon',
220     };
221     res.writeHead(200, {
222         'Content-Type': mime[extname(file)] || 'application/octet-stream',
223         'Cache-Control': 'no-cache',
224     });
225     res.end(req.method === 'HEAD' ? undefined : content);
226 } catch (error) {
227     const status =
228         error instanceof HttpError
229             ? error.status
230             : error instanceof InputError
231                 ? 400
232                 : 500;
233     if (status === 500) console.error('实验数据服务异常: ', error);
234     if (!res.headersSent)
235         json(res, status, {
236             error:
237                 status === 500
238                     ? '读取或保存记录失败: 原文件已保留, 请检查数据目录或恢复备份。'
239                     : (error as Error).message,
240         });
241     else res.end();
242 }
243 });
244 }
245 import { randomUUID } from 'node:crypto';
246 import { STEP_NAMES, STEP_POINTS, WEIGHTS } from '../shared/types.ts';
247 import type { Action, Session } from '../shared/types.ts';
248 export class InputError extends Error {}
249 export function createSession(
250     label = '天平实验练习',
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```
251   source: 'practice' | 'demo' = 'practice',
252 ): Session {
253   const now = new Date().toISOString();
254   return {
255     id: randomUUID(),
256     label,
257     source,
258     createdAt: now,
259     updatedAt: now,
260     finishedAt: null,
261     status: 'active',
262     checked: false,
263     calibrated: false,
264     rider: 10,
265     nut: 2,
266     objectSide: 'box',
267     objectMass: 374,
268     weights: [],
269     tool: 'tweezers',
270     steps: STEP_NAMES.map((name, i) => ({
271       name,
272       max: STEP_POINTS[i],
273       passed: false,
274       score: 0,
275     })),
276     events: [],
277     reading: null,
278     score: 0,
279     feedback: '请先检查器材，再将游码归零。',
280     balanced: false,
281     difference: 10,
282   };
283 }
284 function integer(value: unknown, min: number, max: number): value is number {
285   return Number.isInteger(value) && Number(value) >= min && Number(value) <= max;
286 }
287 export function validateAction(value: unknown): asserts value is Action {
288   if (!value || typeof value !== 'object')
289     throw new InputError('操作必须是 JSON 对象。');
290   const a = value as Record<string, unknown>;
291   if (typeof a.type !== 'string') throw new InputError('操作类型必须是字符串。');
292   const valid =
293     ['check', 'tidy'].includes(a.type) ||
294     (a.type === 'rider' && integer(a.value, 0, 50)) ||
295     (a.type === 'nut' && integer(a.value, -5, 5)) ||
296     (a.type === 'tool' && typeof a.value === 'string' && ['tweezers', 'hand'].includes(a.value)) ||
297     (a.type === 'object' && typeof a.side === 'string' && ['box', 'left', 'right'].includes(a.side)) ||
298     (a.type === 'addWeight' &&
299       WEIGHTS.includes(Number(a.mass)) &&
300       typeof a.mass === 'number' &&
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301     typeof a.side === 'string' && ['left', 'right'].includes(a.side)) ||
302     (a.type === 'removeWeight' && integer(a.index, 0, 3)) ||
303     (a.type === 'read' && integer(a.value, 0, 10000));
304     if (!valid) throw new InputError('操作类型或参数无效。质量参数使用整数十分之一克。');
305 }
306 function pass(s: Session, index: number) {
307     if (s.steps.slice(0, index).every((step) => step.passed)) {
308         s.steps[index].passed = true;
309         s.steps[index].score = s.steps[index].max;
310     }
311 }
312 function empty(s: Session) {
313     return s.objectSide === 'box' && s.weights.length === 0;
314 }
315 function measure(s: Session) {
316     const left =
317         (s.objectSide === 'left' ? s.objectMass : 0) +
318         s.weights.filter((w) => w.side === 'left').reduce((n, w) => n + w.mass, 0);
319     const right =
320         (s.objectSide === 'right' ? s.objectMass : 0) +
321         s.weights.filter((w) => w.side === 'right').reduce((n, w) => n + w.mass, 0);
322     s.difference = left - right - s.rider + s.nut * 10;
323     s.balanced = s.difference === 0;
324 }
325 function properWeights(s: Session) {
326     return (
327         s.weights.length > 0 &&
328         s.weights.every((w) => {
329             const event = [...s.events]
330                 .reverse()
331                 .find((e) => e.action.startsWith(`addWeight:${w.mass}:`));
332             return w.side === 'right' && event?.action.endsWith(':valid');
333         })
334     );
335 }
336 function readyToRead(s: Session) {
337     return (
338         s.calibrated &&
339         s.nut === 0 &&
340         s.objectSide === 'left' &&
341         properWeights(s) &&
342         s.balanced &&
343         s.steps[5].passed
344     );
345 }
346 export function applyAction(previous: Session, action: Action): Session {
347     validateAction(action);
348     if (previous.status !== 'active') throw new InputError('实验已结束，请新建实验。');
349     const s = structuredClone(previous);
350     let violation = false;
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351 let message = '';
352 let tag: string = action.type;
353 const reject = (text: string) => {
354   violation = true;
355   message = text;
356 };
357 switch (action.type) {
358   case 'check':
359     s.checked = true;
360     pass(s, 0);
361     message = '器材检查完成：天平、物体、砝码、镊子齐全。';
362     break;
363   case 'rider':
364     s.rider = action.value;
365     message = `游码已移至 ${s.rider / 10}.toFixed(1)} g。`;
366     if (!s.checked) reject('请先检查器材。');
367     else if (s.rider === 0 && empty(s)) pass(s, 1);
368     else if (!s.calibrated) reject('请将游码归零并完成空载调平。');
369     break;
370   case 'nut':
371     s.nut = action.value;
372     s.calibrated = false;
373     message = `平衡螺母设置为 ${s.nut}。`;
374     if (!empty(s))
375       reject('有负载时禁止调节平衡螺母；请取下物体和砝码、游码归零后重新调平。');
376     else if (!s.checked || !s.steps[1].passed || s.rider !== 0)
377       reject('调平前请先检查器材并将游码归零。');
378     else if (s.nut === 0) {
379       s.calibrated = true;
380       pass(s, 2);
381       message = '空载调平完成。';
382     }
383     break;
384   case 'tool':
385     s.tool = action.value;
386     message =
387       s.tool === 'tweezers'
388       ? '已选用镊子。'
389       : '已切换为徒手操作；取放砝码应使用镊子。';
390     break;
391   case 'object':
392     s.objectSide = action.side;
393     message =
394       action.side === 'box'
395       ? '物体已放回器材区。'
396       : `物体已放在 ${action.side === 'left' ? '左' : '右'} 盘。`;
397     if (action.side === 'right') reject('应左物右码，请将物体放在左盘。');
398     else if (
399       action.side === 'left' &&
400       (!s.calibrated || s.nut !== 0 || s.rider !== 0)
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401     )
402     reject('放物前请游码归零，并完成空载调平。');
403     else if (action.side === 'left') pass(s, 3);
404     break;
405 case 'addWeight': {
406     if (s.weights.some((w) => w.mass === action.mass))
407         throw new InputError('该砝码已在托盘中。');
408     const last = s.weights.at(-1);
409     const valid =
410         s.tool === 'tweezers' &&
411         action.side === 'right' &&
412         s.objectSide === 'left' &&
413         s.calibrated &&
414         s.nut === 0 &&
415         s.steps[3].passed &&
416         (!last || action.mass < last.mass);
417     s.weights.push({ mass: action.mass, side: action.side });
418     tag = `addWeight:${action.mass}:${valid ? 'valid' : 'invalid'}`;
419     message = `已添加 ${action.mass / 10}.toFixed(1)} g 砝码。`;
420     if (!valid)
421         reject(
422             '请先正确调平并左盘放物，使用镊子向右盘按从大到小顺序加码；错误砝码请取下后重新添加。',
423         );
424     else pass(s, 4);
425     break;
426 }
427 case 'removeWeight': {
428     if (!s.weights[action.index]) throw new InputError('砝码索引不存在。');
429     const [removed] = s.weights.splice(action.index, 1);
430     message = `已取回 ${removed.mass / 10}.toFixed(1)} g 砝码。`;
431     if (s.tool !== 'tweezers') reject('取回砝码也应使用镊子。');
432     break;
433 }
434 case 'read':
435     measure(s);
436     if (!readyToRead(s))
437         reject('当前不能读数：需正确调平、左物右码、规范取码，并完成游码微调平衡。');
438     else if (action.value !== s.objectMass)
439         reject('读数不正确，请将右盘砝码总质量与游码示值相加。');
440     else {
441         s.reading = action.value;
442         s.measurement = {
443             weightTotal: s.weights.reduce((n, w) => n + w.mass, 0),
444             rider: s.rider,
445             value: action.value,
446             at: new Date().toISOString(),
447         };
448         pass(s, 6);
449         message = `测量结果 ${s.reading / 10}.toFixed(1)} g，记录成功。`;
450     }
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451     break;
452   case 'tidy':
453     if (!s.steps[6].passed) reject('请先完成正确读数，再整理器材。');
454     else {
455       s.objectSide = 'box';
456       s.weights = [];
457       s.rider = 0;
458       s.nut = 0;
459       s.tool = 'tweezers';
460       s.calibrated = true;
461       pass(s, 7);
462       message = '物体和砝码已归位、游码已归零，实验完成。';
463     }
464     break;
465   }
466   s.updatedAt = new Date().toISOString();
467   s.events.push({ at: s.updatedAt, action: tag, message, violation });
468   measure(s);
469   if (
470     action.type === 'rider' &&
471     !violation &&
472     s.rider > 0 &&
473     s.calibrated &&
474     s.nut === 0 &&
475     s.objectSide === 'left' &&
476     properWeights(s) &&
477     s.balanced
478   ) {
479     pass(s, 5);
480     message = '游码微调完成，天平平衡。请读取右盘砝码与游码的合计质量。';
481     s.events[s.events.length - 1].message = message;
482   }
483   s.score = s.steps.reduce((n, step) => n + step.score, 0);
484   s.feedback = message;
485   return s;
486 }
487 export function finishSession(previous: Session): Session {
488   if (previous.status === 'finished') return previous;
489   const s = structuredClone(previous);
490   s.status = 'finished';
491   s.finishedAt = new Date().toISOString();
492   s.updatedAt = s.finishedAt;
493   s.feedback = `实验已归档，练习完成得分 ${s.score} / 100.`;
494   s.events.push({
495     at: s.finishedAt,
496     action: 'finish',
497     message: s.feedback,
498     violation: false,
499   });
500   return s;
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```
501 }
502 import { resolve } from 'node:path';
503 import { fileURLToPath } from 'node:url';
504 import { createServer } from './app.ts';
505 import { SOFTWARE_NAME, VERSION } from '../shared/types.ts';
506 const root = fileURLToPath(new URL('../', import.meta.url));
507 const port = Number(process.env.PORT || 3187);
508 if (!Number.isInteger(port) || port < 1 || port > 65535)
509   throw new Error('PORT 必须为 1 至 65535 的整数。');
510 const server = createServer(
511   resolve(root, process.env.DATA_DIR || 'data'),
512   resolve(root, 'dist'),
513 );
514 server.on('error', (error: NodeJS.ErrnoException) => {
515   console.error(
516     error.code === 'EADDRINUSE'
517       ? `端口 ${port} 已被占用, 请关闭旧服务或修改 PORT 后重试。`
518       : error.message,
519   );
520   process.exitCode = 1;
521 });
522 server.listen(port, '127.0.0.1', () =>
523   console.log(`${SOFTWARE_NAME} ${VERSION}\n 本机访问: http://127.0.0.1:${port}`),
524 );
525 for (const signal of ['SIGINT', 'SIGTERM'] as const)
526   process.on(signal, () => server.close(() => process.exit(0)));
527 import { mkdir, readFile, rename, writeFile } from 'node:fs/promises';
528 import { join } from 'node:path';
529 import { randomUUID } from 'node:crypto';
530 import { STEP_POINTS, WEIGHTS } from '../shared/types.ts';
531 import type { Session } from '../shared/types.ts';
532 function validSession(value: unknown): value is Session {
533   if (!value || typeof value !== 'object') return false;
534   const s = value as Session;
535   return (
536     typeof s.id === 'string' &&
537     typeof s.label === 'string' &&
538     ['practice', 'demo'].includes(s.source) &&
539     ['active', 'finished'].includes(s.status) &&
540     typeof s.createdAt === 'string' &&
541     typeof s.updatedAt === 'string' &&
542     (s.finishedAt === null || typeof s.finishedAt === 'string') &&
543     typeof s.checked === 'boolean' &&
544     typeof s.calibrated === 'boolean' &&
545     Number.isInteger(s.rider) &&
546     s.rider >= 0 &&
547     s.rider <= 50 &&
548     Number.isInteger(s.nut) &&
549     s.nut >= -5 &&
550     s.nut <= 5 &&

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551 s.objectMass === 374 &&
552 ['box', 'left', 'right'].includes(s.objectSide) &&
553 ['tweezers', 'hand'].includes(s.tool) &&
554 Array.isArray(s.weights) &&
555 s.weights.length <= WEIGHTS.length &&
556 new Set(s.weights.map((w) => w?.mass)).size === s.weights.length &&
557 s.weights.every(
558   (w) => w && WEIGHTS.includes(w.mass) && ['left', 'right'].includes(w.side),
559 ) &&
560 Array.isArray(s.steps) &&
561 s.steps.length === 8 &&
562 s.steps.every(
563   (step, i) =>
564     step &&
565     typeof step.name === 'string' &&
566     typeof step.passed === 'boolean' &&
567     step.max === STEP_POINTS[i] &&
568     step.score === (step.passed ? step.max : 0),
569 ) &&
570 s.score === s.steps.reduce((total, step) => total + step.score, 0) &&
571 Array.isArray(s.events) &&
572 s.events.every(
573   (e) =>
574     e &&
575     typeof e.at === 'string' &&
576     typeof e.action === 'string' &&
577     typeof e.message === 'string' &&
578     typeof e.violation === 'boolean',
579 ) &&
580 (s.reading === null || Number.isInteger(s.reading)) &&
581 typeof s.feedback === 'string' &&
582 typeof s.balanced === 'boolean' &&
583 Number.isInteger(s.difference)
584 );
585 }
586 export class Store {
587   private queue: Promise<unknown> = Promise.resolve();
588   private file: string;
589   private directory: string;
590   constructor(directory: string) {
591     this.directory = directory;
592     this.file = join(directory, 'sessions.json');
593   }
594   private async read(): Promise<Session[]> {
595     let raw: string;
596     try {
597       raw = await readFile(this.file, 'utf8');
598     } catch (error) {
599       if ((error as NodeJS.ErrnoException).code === 'ENOENT') return [];
600       throw error;
601     }
602   }
603 }
```

```
601     }
602     const sessions: unknown = JSON.parse(raw);
603     if (
604       !Array.isArray(sessions) ||
605       !sessions.every(validSession) ||
606       new Set(sessions.map((s) => s.id)).size !== sessions.length
607     ) {
608       throw new Error('记录文件损坏; 请保留原文件并从备份恢复。');
609     }
610     return sessions as Session[];
611   }
612   async list(): Promise<Session[]> {
613     await this.queue;
614     return this.read();
615   }
616   mutate<T>(change: (sessions: Session[]) => T): Promise<T> {
617     const operation = this.queue.then(async () => {
618       const sessions = await this.read();
619       const result = change(sessions);
620       await mkdir(this.directory, { recursive: true });
621       const temporary = `${this.file}.${randomUUID()}.tmp`;
622       await writeFile(temporary, JSON.stringify(sessions, null, 2), {
623         encoding: 'utf8',
624         flag: 'wx',
625       });
626       await rename(temporary, this.file);
627       return result;
628     });
629     this.queue = operation.catch(() => undefined);
630     return operation;
631   }
632 }
633 import React, { useEffect, useState } from 'react';
634 import { createRoot } from 'react-dom/client';
635 import {
636   SOFTWARE_NAME,
637   VERSION,
638   STEP_NAMES,
639   STEP_POINTS,
640   WEIGHTS,
641   type Session,
642   type Action,
643 } from '../shared/types';
644 import './style.css';
645 const grams = (n: number) => (n / 10).toFixed(1);
646 const date = (s: string) => new Date(s).toLocaleString('zh-CN', { hour12: false });
647 async function api<T>(path: string, body?: unknown): Promise<T> {
648   const response = await fetch(
649     '/api' + path,
650     body === undefined
```

```

651     ? {}
652     : {
653         method: 'POST',
654         headers: { 'Content-Type': 'application/json' },
655         body: JSON.stringify(body),
656     },
657 );
658 const data = await response.json();
659 if (!response.ok) throw new Error(data.error || '操作未完成, 请重试');
660 return data;
661 }
662 function Balance({ session: s }: { session: Session }) {
663     const angle = Math.max(-13, Math.min(13, -s.difference / 12));
664     const pan = (side: 'left' | 'right', x: number) => {
665         const y =
666             175 + (side === 'left' ? -1 : 1) * Math.sin((angle * Math.PI) / 180) * 215;
667         return (
668             <g>
669                 <path
670                     d={`M ${x} ${y} L ${x - 66} ${y + 102} M ${x} ${y} L ${x + 66} ${y + 102}`}
671                     stroke="#8b9fa8"
672                     strokeWidth="2"
673                     fill="none"
674                 />
675                 <path
676                     d={`M ${x - 78} ${y + 102} Q ${x - 71} ${y + 123} ${x} ${y + 125} Q ${x + 71} ${y + 123} ${x +
677                     78} ${y + 102} Z`}
678                     fill="url(#steel)"
679                     stroke="#71909e"
680                 />
681                 {s.objectSide === side && (
682                     <g>
683                         <path
684                             d={`M ${x - 25} ${y + 54} l 38 -10 20 14 -1 43 -45 0 -13 -12 Z`}
685                             fill="#bb7658"
686                             stroke="#925339"
687                         />
688                         <path
689                             d={`M ${x - 25} ${y + 54} l 20 12 38 -8 M ${x - 5} ${y + 66} v 35`}
690                             fill="none"
691                             stroke="#e4ac87"
692                             strokeWidth="2"
693                         />
694                     </g>
695                 )}
696             <g>
697                 {s.weights
698                     .filter((w) => w.side === side)
699                     .map((w, i) => (
700                         <g>
701                             key={w.mass}
702                             transform={`translate(${x - 48 + i * 32}, ${y + 81 - (w.mass === 500 ? 12 : 0)})`}

```

```

701     >
702     <rect x="7" y="-7" width="12" height="8" rx="3" fill="#8e7041" />
703     <path
704     d="M 6 0 L 20 0 L 25 20 L 1 20 Z"
705     fill="url(#brass)"
706     stroke="#957645"
707     />
708     <text x="13" y="14" textAnchor="middle" fontSize="9" fill="#4d3d25">
709     {w.mass / 10}
710     </text>
711   </g>
712   )))
713   <text x={x} y="335" textAnchor="middle" className="pan-label">
714   {side === 'left' ? '左盘 · 放置物体' : '右盘 · 添加砝码'}
715   </text>
716 </g>
717 );
718 };
719 return (
720   <svg
721     className="balance"
722     viewBox="0 0 820 390"
723     role="img"
724     aria-label={`托盘天平, ${s.balanced ? '已平衡' : '未平衡'}, 左盘${s.objectSide === 'left' ? '有物体' : '无物体'}`}
725   >
726     <defs>
727       <linearGradient id="steel" x1="0" y1="0" x2="0" y2="1">
728         <stop stopColor="#ecf4f5" />
729         <stop offset=".6" stopColor="#b9cbd2" />
730         <stop offset="1" stopColor="#7c99a7" />
731       </linearGradient>
732       <linearGradient id="brass">
733         <stop stopColor="#d5bb81" />
734         <stop offset=".5" stopColor="#eee0b4" />
735         <stop offset="1" stopColor="#ab8c4c" />
736       </linearGradient>
737       <linearGradient id="base">
738         <stop stopColor="#265468" />
739         <stop offset="1" stopColor="#153748" />
740       </linearGradient>
741     </defs>
742     <ellipse cx="410" cy="364" rx="218" ry="10" fill="#173d4c" opacity=".07" />
743     <path d="M 332 349 L 346 327 H 474 L 490 349 Z" fill="url(#base)" />
744     <rect x="393" y="172" width="34" height="155" rx="4" fill="url(#steel)" />
745     <path d="M 379 327 L 398 282 H 422 L 441 327" fill="#27556b" />
746     <path d="M 357 177 A 56 56 0 0 1 463 177" fill="fff" stroke="#b9cbd2" />
747     {[ -40, -20, 0, 20, 40 ].map((v) => (
748       <line
749         key={v}
750         x1={410 + v}

```

```

751     y1={v === 0 ? 124 : 130 + Math.abs(v) / 4}
752     x2={410 + v}
753     y2={v === 0 ? 138 : 139 + Math.abs(v) / 4}
754     stroke={v === 0 ? '#0d887f' : '#8ba2ad'}
755     strokeWidth={v === 0 ? 3 : 1}
756   />
757 )}}
758 <g transform={`rotate(${angle},410,175)`} className="beam">
759   <path
760     d="M 193 173 L 410 161 L 627 173 L 627 185 L 410 180 L 193 185 Z"
761     fill="url(#steel)"
762     stroke="#73909e"
763   />
764   <line x1="410" y1="171" x2="410" y2="124" stroke="#148d85" strokeWidth="3" />
765   <circle cx="410" cy="175" r="9" fill="#31586a" />
766   <circle cx="410" cy="175" r="3" fill="#e5eef1" />
767   <rect x="624" y="167" width="18" height="24" rx="3" fill="url(#brass)" />
768   <rect x="180" y="167" width="18" height="24" rx="3" fill="url(#brass)" />
769   <rect
770     x="426"
771     y="186"
772     width="181"
773     height="15"
774     rx="2"
775     fill="#dce7eb"
776     stroke="#8ea7b2"
777   />
778   {Array.from({ length: 26 }, (_, i) => (
779     <line
780       key={i}
781       x1={430 + i * 6.8}
782       x2={430 + i * 6.8}
783       y1="187"
784       y2={i % 5 === 0 ? 197 : 192}
785       stroke="#637d89"
786     />
787   )}}
788   <rect
789     x={426 + s.rider * 3.4}
790     y="182"
791     width="9"
792     height="25"
793     rx="2"
794     fill="#187e7b"
795   />
796 </g>
797 {pan('left', 195)}
798 {pan('right', 625)}
799 <text
800   x="410"

```

```
801     y="349"
802     textAnchor="middle"
803     fill="#d8e7eb"
804     fontSize="10"
805     letterSpacing="3"
806   >
807     TRAY BALANCE
808   </text>
809   <g transform="translate(350,35)">
810     <rect
811       width="120"
812       height="28"
813       rx="14"
814       fill={s.balanced ? '#d8eee7' : '#e1e9ee'}
815     />
816     <circle cx="17" cy="14" r="3" fill={s.balanced ? '#168778' : '#617a89'} />
817     <text x="67" y="19" textAnchor="middle" fill="#285365" fontSize="12">
818       {s.balanced ? '指针已平衡' : '指针未平衡'}
819     </text>
820   </g>
821 </svg>
822 );
823 }
824 function App() {
825   const [tab, setTab] = useState('实验台'),
826     [session, setSession] = useState<Session | null>(null),
827     [history, setHistory] = useState<Session[]>([]),
828     [busy, setBusy] = useState(false),
829     [error, setError] = useState(''),
830     [label, setLabel] = useState('天平质量测量练习'),
831     [source, setSource] = useState<'practice' | 'demo'>('practice'),
832     [side, setSide] = useState<'left' | 'right'>('right'),
833     [reading, setReading] = useState(''),
834     [report, setReport] = useState<Session | null>(null),
835     [reset, setReset] = useState(false),
836     [online, setOnline] = useState(false);
837   const run = async (fn: () => Promise<void>) => {
838     setBusy(true);
839     setError('');
840     try {
841       await fn();
842     } catch (e) {
843       setError(e instanceof Error ? e.message : '连接失败, 请检查本地服务后重试');
844     } finally {
845       setBusy(false);
846     }
847   };
848   const load = () =>
849     run(async () => {
850       await api('/health');
```

```
851     setOnline(true);
852     setHistory(await api<Session[]>('/sessions'));
853     const id = localStorage.getItem('balance-session');
854     if (id) setSession(await api<Session>('/sessions/' + id));
855   });
856   useEffect(() => {
857     void load();
858   }, []);
859   useEffect(() => {
860     if (!reset && !report) return;
861     const dialog = document.querySelector<HTMLElement>(
862       report ? '.report[role="dialog"]' : '.new-dialog[role="dialog"]',
863     );
864     if (!dialog) return;
865     const previousFocus = document.activeElement instanceof HTMLElement
866       ? document.activeElement : null;
867     const focusable = () => Array.from(dialog.querySelectorAll<HTMLElement>(
868       'button:not(:disabled), input:not(:disabled), select:not(:disabled), a[href],
869       [tabindex]:not([tabindex="-1"])',
870     )).filter((element) => element.getClientRects().length > 0);
871     dialog.tabIndex = -1;
872     const focusFirst = () => (focusable()[0] ?? dialog).focus();
873     focusFirst();
874     const onKeyDown = (event: KeyboardEvent) => {
875       if (event.key === 'Escape') {
876         event.preventDefault();
877         if (report) setReport(null);
878         else setReset(false);
879       } else if (event.key === 'Tab') {
880         const items = focusable();
881         const first = items[0], last = items[items.length - 1];
882         if (!first) {
883           event.preventDefault();
884           dialog.focus();
885         } else if (event.shiftKey &&
886           (document.activeElement === first || !dialog.contains(document.activeElement))) {
887           event.preventDefault();
888           last.focus();
889         } else if (!event.shiftKey &&
890           (document.activeElement === last || !dialog.contains(document.activeElement))) {
891           event.preventDefault();
892           first.focus();
893         }
894       }
895     };
896     const onFocusIn = (event: FocusEvent) => {
897       if (event.target instanceof Node && !dialog.contains(event.target)) focusFirst();
898     };
899     document.addEventListener('keydown', onKeyDown);
900     document.addEventListener('focusin', onFocusIn);
901     return () => {
```

```
901     document.removeEventListener('keydown', onKeyDown);
902     document.removeEventListener('focusin', onFocusIn);
903     if (previousFocus?.isConnected) previousFocus.focus();
904   };
905 }, [reset, report?.id]);
906 const create = () =>
907   run(async () => {
908     const s = await api<Session>('/sessions', {
909       label: label.trim() || '天平质量测量练习',
910       source,
911     });
912     setSession(s);
913     localStorage.setItem('balance-session', s.id);
914     setReading('');
915     setSide('right');
916     setReset(false);
917     setTab('实验台');
918     setHistory(await api('/sessions'));
919   });
920 const act = (action: Action) =>
921   session &&
922   run(async () =>
923     setSession(await api<Session>(`/sessions/${session.id}/actions`, action)),
924   );
925 const finish = () =>
926   session &&
927   run(async () => {
928     const s = await api<Session>(`/sessions/${session.id}/finish`, {});
929     setSession(s);
930     setReport(s);
931     setHistory(await api('/sessions'));
932   });
933 const disabled = busy || session?.status === 'finished';
934 const submitReading = () => {
935   if (
936     !/^\\d+(\\.\\d)?$/ .test(reading) ||
937     Number(reading) < 0 ||
938     Number(reading) > 1000
939   ) {
940     setError('读数需为 0 至 1000 g 的数字, 最多保留一位小数。');
941     return;
942   }
943   void act({ type: 'read', value: Math.round(Number(reading) * 10) });
944 };
945 const download = (s: Session) => {
946   const a = document.createElement('a');
947   a.href = `/api/sessions/${s.id}/export`;
948   a.download = `天平实验-${s.id}.json`;
949   a.click();
950 };
```

```

951 return (
952   <>
953     <header className="topbar">
954       <div className="brand-icon"><img alt="brand icon" data-bbox="115 105 135 125"/></div>
955       <div className="brand">
956         <strong>{SOFTWARE_NAME}</strong>
957         <span> 实验教学 · 交互练习与过程评测</span>
958       </div>
959       <span className="version">{VERSION}</span>
960       <div className="connection">
961         <i className={online ? 'on' : ''} />
962         {online ? '本地服务已连接' : '正在连接本地服务'}
963       </div>
964     </header>
965     <div className="navigation">
966       <nav aria-label="主导航">
967         {[ '实验台', '实验记录', '实验指南' ].map((t, i) => (
968           <button
969             key={t}
970             className={tab === t ? 'selected' : ''}
971             onClick={() => {
972               setTab(t);
973               if (i === 1) void run(async () => setHistory(await api('/sessions')));
974             }}
975           >
976             <span>{['◆', '▣', '◎'][i]}</span>
977             {t}
978           </button>
979         ))}
980       </nav>
981       <span> 认识仪器，规范操作，准确测量。</span>
982     </div>
983     <main>
984       {error && (
985         <div role="alert" className="error">
986           {error}
987           <button onClick={() => void load()}> 重新连接</button>
988         </div>
989       )}
990       {tab === '实验台' && (
991         <>
992           <div className="page-heading">
993             <div>
994               <div className="eyebrow"> 物理实验 / 质量测量</div>
995               <h1> 把每一步，放在天平上。</h1>
996               <p> 按顺序完成八个实验环节，操作过程将自动保存。</p>
997             </div>
998             <button className="secondary" onClick={() => setReset(true)}>
999               + 新建实验
1000             </button>

```

```

1001     </div>
1002     {!session ? (
1003       <section className="start-panel">
1004         <div className="start-mark">⚖️</div>
1005         <h2> 开始你的第一轮测量</h2>
1006         <p> 检查器材、调整平衡，再测量物体的质量。</p>
1007         <label>
1008           实验名称
1009           <input
1010             value={label}
1011             onChange={(e) => setLabel(e.target.value)}
1012             maxLength={80}
1013           />
1014         </label>
1015         <label>
1016           记录来源
1017           <select
1018             value={source}
1019             onChange={(e) => setSource(e.target.value as 'practice' | 'demo')}
1020           >
1021             <option value="practice"> 自主练习</option>
1022             <option value="demo"> 演示记录</option>
1023           </select>
1024         </label>
1025         <button
1026           className="primary"
1027           disabled={busy}
1028           onClick={() => void create()}
1029         >
1030           开始实验 →
1031         </button>
1032       </section>
1033     ) : (
1034       <div className="workbench">
1035         <div className="left-column">
1036           <section className="instrument-panel">
1037             <div className="panel-heading">
1038               <div>
1039                 <strong>{session.label}</strong>
1040                 <span className="badge">
1041                   {session.source === 'demo' ? '演示记录' : '自主练习'}
1042                 </span>
1043                 {session.status === 'finished' && (
1044                   <span className="badge"> 已归档</span>
1045                 )}
1046               </div>
1047               <span className="subtle"> 分度值 0.1 g</span>
1048             </div>
1049             <Balance session={session} />
1050             <div className="instrument-readings">

```

```

1051     <div>
1052       <span> 游码示值</span>
1053       <strong>
1054         {grams(session.rider)}
1055         <small>g</small>
1056       </strong>
1057     </div>
1058     <div>
1059       <span> 砝码总质量</span>
1060       <strong>
1061         {grams(session.weights.reduce((n, w) => n + w.mass, 0))}
1062         <small>g</small>
1063       </strong>
1064     </div>
1065     <div>
1066       <span> 空载调平</span>
1067       <strong className="status-word">
1068         {session.calibrated ? '已完成' : '待调整'}
1069       </strong>
1070     </div>
1071   </div>
1072 </section>
1073 <div
1074   className={`feedback ${session.events.at(-1)?.violation ? 'warning' : ''}`}
1075   role="status"
1076 >
1077   <span>{session.events.at(-1)?.violation ? '!' : 'i'}</span>
1078   <div>
1079     <strong>
1080       {session.events.at(-1)?.violation ? '操作提示' : '当前指引'}
1081     </strong>
1082     <p>{session.feedback}</p>
1083   </div>
1084 </div>
1085 <section className="controls">
1086   <div className="panel-heading">
1087     <strong> 器材与操作</strong>
1088     <span className="subtle"> 依次完成准备、测量与整理</span>
1089   </div>
1090   <fieldset disabled={disabled}>
1091     <div className="control-grid">
1092       <div className="control-group">
1093         <h3>
1094           <b>01</b> 准备与调平
1095         </h3>
1096         <button
1097           className={session.checked ? 'done-button' : 'secondary'}
1098           onClick={() => void act({ type: 'check' })}
1099         >
1100           {session.checked ? '✓ 器材检查完成' : '检查实验器材'}

```

```
1101     </button>
1102     <label className="slider-label">
1103       平衡螺母 <output>{session.nut}</output>
1104       <input
1105         aria-label=" 平衡螺母"
1106         type="range"
1107         min="-5"
1108         max="5"
1109         value={session.nut}
1110         onChange={(e) =>
1111           void act({ type: 'nut', value: Number(e.target.value) })
1112         }
1113       />
1114     </label>
1115     <small> 空载时, 先游码归零, 再调螺母至 0。</small>
1116   </div>
1117   <div className="control-group">
1118     <h3>
1119       <b>02</b> 放物与取码
1120     </h3>
1121     <label>
1122       取码工具
1123       <select
1124         aria-label=" 取码工具"
1125         value={session.tool}
1126         onChange={(e) =>
1127           void act({
1128             type: 'tool',
1129             value: e.target.value as 'hand' | 'tweezers',
1130           })
1131         }
1132       >
1133         <option value="tweezers"> 镊子</option>
1134         <option value="hand"> 徒手 (不规范操作)</option>
1135       </select>
1136     </label>
1137     <label>
1138       物体位置
1139       <select
1140         aria-label=" 物体位置"
1141         value={session.objectSide}
1142         onChange={(e) =>
1143           void act({
1144             type: 'object',
1145             side: e.target.value as 'box' | 'left' | 'right',
1146           })
1147         }
1148       >
1149         <option value="box"> 器材盒</option>
1150         <option value="left"> 左盘</option>
```

```

1151         <option value="right"> 右盘</option>
1152     </select>
1153 </label>
1154 <label>
1155     砝码放置
1156     <select
1157         aria-label=" 砝码放置"
1158         value={side}
1159         onChange={(e) =>
1160             setSide(e.target.value as 'left' | 'right')
1161         }
1162     >
1163         <option value="right"> 右盘</option>
1164         <option value="left"> 左盘</option>
1165     </select>
1166 </label>
1167 </div>
1168 <div className="control-group">
1169     <h3>
1170         <b>03</b> 砝码与游码
1171     </h3>
1172     <div className="weight-buttons">
1173         {WEIGHTS.map((w) => (
1174             <button
1175                 key={w}
1176                 disabled={session.weights.some((x) => x.mass === w)}
1177                 onClick={() =>
1178                     void act({ type: 'addWeight', mass: w, side })
1179                 }
1180                 aria-label={`添加 ${w / 10} 克砝码`}
1181             >
1182                 <span>■</span>
1183                 {w / 10}
1184                 <small>g</small>
1185             </button>
1186         ))}
1187     </div>
1188     <div className="placed-weights">
1189         {session.weights.length ? (
1190             session.weights.map((w, i) => (
1191                 <button
1192                     key={w.mass}
1193                     onClick={() =>
1194                         void act({ type: 'removeWeight', index: i })
1195                     }
1196                     aria-label={`取回 ${w.mass / 10} 克砝码`}
1197                 >
1198                     {w.mass / 10}g {w.side === 'left' ? '左' : '右'} ×
1199                 </button>
1200             ))

```

```
1201         ) : (  
1202             <span> 砝码均在器材盒内</span>  
1203         )}  
1204     </div>  
1205     <label className="slider-label">  
1206         游码 <output>{grams(session.rider)} g</output>  
1207         <input  
1208             aria-label=" 游码"  
1209             type="range"  
1210             min="0"  
1211             max="50"  
1212             value={session.rider}  
1213             onChange={(e) =>  
1214                 void act({  
1215                     type: 'rider',  
1216                     value: Number(e.target.value),  
1217                 })  
1218             }  
1219         />  
1220     </label>  
1221     <button  
1222         className="text-button"  
1223         onClick={() => void act({ type: 'rider', value: 0 })}  
1224     >  
1225         游码归零  
1226     </button>  
1227 </div>  
1228 </div>  
1229 <div className="record-row">  
1230     <label htmlFor="reading"> 测量读数</label>  
1231     <div className="unit-input">  
1232         <input  
1233             id="reading"  
1234             aria-label=" 测量读数"  
1235             type="number"  
1236             min="0"  
1237             max="1000"  
1238             step="0.1"  
1239             placeholder="0.0"  
1240             value={reading}  
1241             onChange={(e) => setReading(e.target.value)}  
1242         />  
1243         <span>g</span>  
1244     </div>  
1245     <button  
1246         className="primary"  
1247         disabled={reading === '' || !Number.isFinite(Number(reading))}  
1248         onClick={submitReading}  
1249     >  
1250         记录读数
```

```
1251         </button>
1252         <button
1253             className="secondary"
1254             onClick={() => void act({ type: 'tidy' })}
1255         >
1256             整理器材归位
1257         </button>
1258     </div>
1259 </fieldset>
1260 </section>
1261 </div>
1262 <aside>
1263     <section className="assessment">
1264         <div className="panel-heading">
1265             <strong> 实验完成度</strong>
1266             <span className="subtle">8 个环节</span>
1267         </div>
1268         <div className="score">
1269             <strong>{session.score}</strong>
1270             <span>/ 100 分</span>
1271             <p> 练习完成评分</p>
1272         </div>
1273         <div className="progress">
1274             <div style={{ width: session.score + '%' }} />
1275         </div>
1276         <ol className="step-list">
1277             {session.steps.map((s, i) => (
1278                 <li key={s.name} className={s.passed ? 'passed' : ''}>
1279                     <span className="step-no">
1280                         {s.passed ? '✓' : String(i + 1).padStart(2, '0')}
1281                     </span>
1282                     <span>{s.name}</span>
1283                     <b>
1284                         {s.score}
1285                         <small>/ {s.max}</small>
1286                     </b>
1287                 </li>
1288             ))}
1289         </ol>
1290         <p className="assessment-note">
1291             操作失误会记录提示：纠正后仍可完成环节得分。每项只计分一次。
1292         </p>
1293         <button
1294             className="primary full"
1295             disabled={busy}
1296             onClick={() =>
1297                 session.status === 'finished'
1298                 ? setReport(session)
1299                 : void finish()
1300             }

```

```
1301         >
1302         {session.status === 'finished'
1303           ? '查看实验报告'
1304           : '结束实验 • 生成报告'}
1305       </button>
1306     </section>
1307     <section className="event-panel">
1308       <div className="panel-heading">
1309         <strong> 最近操作</strong>
1310         <span className="subtle"> 自动保存</span>
1311       </div>
1312       {session.events
1313         .slice(-3)
1314         .reverse()
1315         .map((e, i) => (
1316           <div className="event" key={i}>
1317             <i className={e.violation ? 'warn' : ''} />
1318             <p>
1319               {e.message}
1320               <time>
1321                 {new Date(e.at).toLocaleTimeString('zh-CN', {
1322                   hour12: false,
1323                 })}
1324             </time>
1325             </p>
1326           </div>
1327         ))}
1328       {!session.events.length && (
1329         <p className="subtle"> 从检查实验器材开始。</p>
1330       )}
1331     </section>
1332   </aside>
1333 </div>
1334 ))}
1335 </>
1336 )}
1337 {tab === '实验记录' && (
1338   <>
1339     <div className="page-heading">
1340       <div className="eyebrow"> 实验档案 / 自动保存</div>
1341       <h1> 每一次练习，都有迹可循。</h1>
1342       <p> 查看测量结果、操作过程与分项完成情况。</p>
1343     </div>
1344     <section className="history-panel">
1345       <table>
1346         <thead>
1347           <tr>
1348             <th> 实验名称</th>
1349             <th> 创建时间</th>
1350             <th> 来源</th>
```

```

1351         <th> 状态</th>
1352         <th> 完成评分</th>
1353         <th> 操作</th>
1354     </tr>
1355 </thead>
1356 <tbody>
1357     {history.map((s) => (
1358         <tr key={s.id}>
1359             <td>
1360                 <strong>{s.label}</strong>
1361                 <small>{s.id.slice(0, 8)}</small>
1362             </td>
1363             <td>{date(s.createdAt)}</td>
1364             <td>{s.source === 'demo' ? '演示记录' : '自主学习'}</td>
1365             <td>
1366                 <span className="badge">
1367                     {s.status === 'finished' ? '已归档' : '进行中'}
1368                 </span>
1369             </td>
1370             <td>
1371                 <b>{s.score}</b> / 100
1372             </td>
1373             <td>
1374                 <button className="text-button" onClick={() => setReport(s)}>
1375                     查看报告
1376                 </button>
1377                 <button className="text-button" onClick={() => download(s)}>
1378                     导出 JSON
1379                 </button>
1380             </td>
1381         </tr>
1382     )})
1383 </tbody>
1384 </table>
1385 {!history.length && (
1386     <div className="empty"> 还没有实验记录。前往实验台开始练习。</div>
1387 )}
1388 </section>
1389 </>
1390 )}
1391 {tab === '实验指南' && (
1392     <>
1393         <div className="page-heading">
1394             <div className="eyebrow"> 操作规范 / 学习指南</div>
1395             <h1> 先调平，再测量。</h1>
1396             <p> 这是一套本地交互练习系统，八个环节共 100 分。</p>
1397         </div>
1398         <div className="guide-grid">
1399             {STEP_NAMES.map((name, i) => (
1400                 <section key={name}>

```

```

1401         <span className="guide-no">{String(i + 1).padStart(2, '0')}</span>
1402         <h2>
1403             {name}
1404             <small>{STEP_POINTS[i]} 分</small>
1405         </h2>
1406         <p>
1407             {
1408                 [
1409                     '先检查天平、砝码、镊子及待测物体，确认器材齐全。',
1410                     '保持托盘空载，将游码移动到标尺的零刻度。',
1411                     '游码归零后，将平衡螺母调整到 0，使空载天平平衡。',
1412                     '完成调平后，将待测物体放在左盘，砝码放在右盘。',
1413                     '用镊子先试较大的砝码，过重则取回，再依次试较小的砝码。',
1414                     '砝码接近物体质量后，微调游码；指针居中即表示达到平衡。',
1415                     '物体质量等于右盘砝码质量加上游码示值，按 0.1 g 分度记录。',
1416                     '记录正确读数后整理器材：物体、砝码归盒，游码与螺母回零。',
1417                 ] [i]
1418             }
1419         </p>
1420     </section>
1421   )})
1422 </div>
1423 <p className="guide-disclaimer">
1424   评分用于练习完成情况反馈，不能替代真实实验考核。错误操作保留在报告中，纠正后可继续完成练习。
1425 </p>
1426 </>
1427 )}
1428 </main>
1429 <footer>
1430   <span>
1431     {SOFTWARE_NAME} {VERSION}
1432   </span>
1433   <span> 本地运行 · 过程留痕 · 自主练习</span>
1434 </footer>
1435 {reset && (
1436   <div className="modal-backdrop">
1437     <section
1438       className="new-dialog"
1439       role="dialog"
1440       aria-modal="true"
1441       aria-labelledby="new-title"
1442     >
1443       <h2 id="new-title"> 新建一轮实验</h2>
1444       <p> 当前记录会保留在实验记录中，新实验从初始状态开始。</p>
1445       <label>
1446         实验名称
1447         <input
1448           value={label}
1449           onChange={(e) => setLabel(e.target.value)}
1450           maxLength={80}

```

```
1451     />
1452   </label>
1453   <label>
1454     记录来源
1455     <select
1456       value={source}
1457       onChange={(e) => setSource(e.target.value as 'practice' | 'demo')}
1458     >
1459       <option value="practice"> 自主练习</option>
1460       <option value="demo"> 演示记录</option>
1461     </select>
1462   </label>
1463   <div className="dialog-actions">
1464     <button className="secondary" onClick={() => setReset(false)}>
1465       取消
1466     </button>
1467     <button className="primary" disabled={busy} onClick={() => void create()}>
1468       开始新实验
1469     </button>
1470   </div>
1471 </section>
1472 </div>
1473 )}
1474 {report && (
1475   <div className="modal-backdrop report-backdrop">
1476     <section
1477       className="report"
1478       role="dialog"
1479       aria-modal="true"
1480       aria-labelledby="report-title"
1481     >
1482       <div className="report-tools">
1483         <button className="secondary" onClick={() => download(report)}>
1484           导出 JSON
1485         </button>
1486         <button className="primary" onClick={() => window.print()}>
1487           打印 / 保存 PDF
1488         </button>
1489         <button className="secondary" onClick={() => setReport(null)}>
1490           关闭报告
1491         </button>
1492       </div>
1493       <div className="eyebrow">
1494         实验评测报告 • {report.source === 'demo' ? '演示记录' : '自主练习'}
1495       </div>
1496       <h1 id="report-title">{report.label}</h1>
1497       <p>
1498         {SOFTWARE_NAME} {VERSION}
1499       </p>
1500       <div className="report-summary">
```

```

1501     <div>
1502         <span> 完成评分</span>
1503         <strong>{report.score} / 100</strong>
1504     </div>
1505     <div>
1506         <span> 记录读数</span>
1507         <strong>
1508             {report.reading === null ? '尚未记录' : grams(report.reading) + ' g'}
1509         </strong>
1510     </div>
1511     <div>
1512         <span> 状态</span>
1513         <strong>{report.status === 'finished' ? '已归档' : '进行中'}</strong>
1514     </div>
1515 </div>
1516 <p className="subtle">
1517     创建时间: {date(report.createdAt)}
1518     <br />
1519     结束时间: {report.finishedAt ? date(report.finishedAt) : '尚未结束'}
1520     <br />
1521     记录编号: {report.id}
1522 </p>
1523 <h2> 测量结果</h2>
1524 <p>
1525     {report.measurement
1526     ? `右盘砝码 ${grams(report.measurement.weightTotal)} g + 游码
1527       ${grams(report.measurement.rider)} g = 物体质量 ${grams(report.measurement.value)} g`
1528     : '尚未完成正确读数。'}
1529 </p>
1530 <h2> 分项完成情况</h2>
1531 <table>
1532     <thead>
1533         <tr>
1534             <th> 实验环节</th>
1535             <th> 状态</th>
1536             <th> 得分</th>
1537         </tr>
1538     </thead>
1539     <tbody>
1540         {report.steps.map((s) => (
1541             <tr key={s.name}>
1542                 <td>{s.name}</td>
1543                 <td>{s.passed ? '已完成' : '未完成'}</td>
1544                 <td>
1545                     {s.score} / {s.max}
1546                 </td>
1547             </tr>
1548         ))}
1549     </tbody>
1550 </table>
<h2> 完整操作记录</h2>

```

```
1551         <table className="report-events">
1552             <thead>
1553                 <tr>
1554                     <th> 时间</th>
1555                     <th> 类型</th>
1556                     <th> 操作反馈</th>
1557                 </tr>
1558             </thead>
1559             <tbody>
1560                 {report.events.map((e, i) => (
1561                     <tr key={i}>
1562                         <td>{date(e.at)}</td>
1563                         <td>{e.violation ? '违规提示' : '操作记录'}</td>
1564                         <td>{e.message}</td>
1565                     </tr>
1566                 ))}
1567             </tbody>
1568         </table>
1569         <p className="report-note">
1570             本报告为本地交互练习的完成评分。失误记录保留，纠正后允许获得环节得分；报告不代替真实实验考核。
1571         </p>
1572     </section>
1573 </div>
1574     )}
1575 </>
1576 );
1577 }
1578 createRoot(document.getElementById('root')).render(
1579     <React.StrictMode>
1580         <App />
1581     </React.StrictMode>,
1582 );
1583 :root {
1584     font-family: 'Microsoft YaHei', 'PingFang SC', sans-serif;
1585     color: #243e4e;
1586     background: #f3f6f8;
1587     font-synthesis: none;
1588     font-size: 14px;
1589     --navy: #163c51;
1590     --teal: #14857d;
1591     --line: #e0e8ec;
1592     --muted: #718592;
1593 }
1594 * {
1595     box-sizing: border-box;
1596 }
1597 body {
1598     margin: 0;
1599 }
1600 button,
```

```
1601 input,
1602 select {
1603   font: inherit;
1604 }
1605 button {
1606   cursor: pointer;
1607   transition:
1608     background 0.15s,
1609     box-shadow 0.15s;
1610 }
1611 button:disabled {
1612   cursor: not-allowed;
1613   opacity: 0.48;
1614 }
1615 button:focus-visible,
1616 input:focus-visible,
1617 select:focus-visible {
1618   outline: 3px solid #55b5bf;
1619   outline-offset: 3px;
1620 }
1621 button {
1622   border: 0;
1623   border-radius: 5px;
1624 }
1625 input,
1626 select {
1627   border: 1px solid #d7e1e6;
1628   background: white;
1629   color: #284555;
1630   border-radius: 4px;
1631   padding: 9px 10px;
1632   min-width: 0;
1633 }
1634 input[type='range'] {
1635   width: 100%;
1636   accent-color: var(--teal);
1637   padding: 0;
1638   height: 20px;
1639   cursor: pointer;
1640 }
1641 input[type='range']:disabled {
1642   cursor: not-allowed;
1643 }
1644 .topbar {
1645   height: 81px;
1646   background: #fff;
1647   display: flex;
1648   align-items: center;
1649   padding: 0 4%;
1650   gap: 14px;
```

```
1651 border-bottom: 1px solid var(--line);
1652 }
1653 .brand-icon {
1654   background: var(--navy);
1655   color: white;
1656   width: 43px;
1657   height: 43px;
1658   display: grid;
1659   place-items: center;
1660   border-radius: 8px;
1661   font-size: 27px;
1662 }
1663 .brand {
1664   display: flex;
1665   flex-direction: column;
1666   gap: 5px;
1667 }
1668 .brand strong {
1669   font-size: 17px;
1670   letter-spacing: 0.5px;
1671 }
1672 .brand span {
1673   color: var(--muted);
1674   font-size: 11px;
1675   letter-spacing: 2px;
1676 }
1677 .version {
1678   font-family: Consolas, monospace;
1679   font-size: 11px;
1680   background: #eef3f6;
1681   color: #758c99;
1682   padding: 4px 7px;
1683   border-radius: 4px;
1684   align-self: center;
1685 }
1686 .connection {
1687   margin-left: auto;
1688   display: flex;
1689   align-items: center;
1690   gap: 7px;
1691   color: #728391;
1692   font-size: 12px;
1693 }
1694 .connection i {
1695   width: 6px;
1696   height: 6px;
1697   background: #a5b1ba;
1698   border-radius: 50%;
1699 }
1700 .connection i.on {
```

```
1701   background: #159583;
1702   box-shadow: 0 0 0 4px #edf7f4;
1703 }
1704 .navigation {
1705   height: 55px;
1706   background: white;
1707   display: flex;
1708   justify-content: space-between;
1709   align-items: center;
1710   padding: 0 4%;
1711   border-bottom: 1px solid var(--line);
1712 }
1713 .navigation nav {
1714   display: flex;
1715   height: 100%;
1716   gap: 30px;
1717 }
1718 .navigation button {
1719   padding: 0 5px;
1720   background: transparent;
1721   border-radius: 0;
1722   color: #748590;
1723   border-bottom: 3px solid transparent;
1724   display: flex;
1725   align-items: center;
1726   gap: 10px;
1727 }
1728 .navigation button span {
1729   font-size: 18px;
1730 }
1731 .navigation button.selected {
1732   color: var(--teal);
1733   border-bottom-color: var(--teal);
1734   font-weight: 700;
1735 }
1736 .navigation > span {
1737   font-size: 11px;
1738   color: #91a1aa;
1739 }
1740 main {
1741   max-width: 1510px;
1742   margin: auto;
1743   padding: 25px 4% 35px;
1744   min-height: calc(100vh - 181px);
1745 }
1746 .page-heading {
1747   position: relative;
1748   margin: 0 0 23px;
1749 }
1750 .page-heading > button {
```

```
1751 position: absolute;
1752 right: 0;
1753 bottom: 2px;
1754 }
1755 .eyebrow {
1756 font-size: 11px;
1757 letter-spacing: 2px;
1758 color: #648995;
1759 font-weight: 600;
1760 margin-bottom: 8px;
1761 }
1762 h1 {
1763 font-size: 26px;
1764 letter-spacing: 1px;
1765 color: #1b3c50;
1766 font-weight: 600;
1767 margin: 0 0 8px;
1768 }
1769 p {
1770 line-height: 1.7;
1771 }
1772 h2 {
1773 font-size: 18px;
1774 }
1775 h3 {
1776 font-size: 12px;
1777 font-weight: 600;
1778 margin: 0 0 14px;
1779 }
1780 .page-heading p {
1781 font-size: 12px;
1782 color: var(--muted);
1783 margin: 0;
1784 }
1785 .primary {
1786 padding: 10px 16px;
1787 background: var(--teal);
1788 color: white;
1789 font-weight: 600;
1790 border: 1px solid var(--teal);
1791 }
1792 .primary:hover:not(:disabled) {
1793 background: #0c716c;
1794 }
1795 .secondary {
1796 background: white;
1797 border: 1px solid #d8e3e8;
1798 padding: 10px 15px;
1799 color: #446372;
1800 }
```

```
1801 .secondary:hover:not(:disabled) {
1802   background: #eef5f7;
1803 }
1804 .text-button {
1805   background: transparent;
1806   color: var(--teal);
1807   padding: 6px;
1808   font-size: 12px;
1809 }
1810 .workbench {
1811   display: grid;
1812   grid-template-columns: minmax(0, 1fr) 310px;
1813   gap: 22px;
1814   align-items: start;
1815 }
1816 .left-column {
1817   min-width: 0;
1818 }
1819 .instrument-panel,
1820 .controls,
1821 .assessment,
1822 .event-panel,
1823 .history-panel,
1824 .guide-grid section {
1825   background: white;
1826   border: 1px solid var(--line);
1827   border-radius: 8px;
1828   overflow: hidden;
1829 }
1830 .panel-heading {
1831   padding: 17px 20px;
1832   display: flex;
1833   align-items: center;
1834   justify-content: space-between;
1835   gap: 10px;
1836 }
1837 .panel-heading strong {
1838   font-size: 13px;
1839   font-weight: 600;
1840 }
1841 .panel-heading > div {
1842   display: flex;
1843   align-items: center;
1844   gap: 9px;
1845 }
1846 .badge {
1847   font-size: 10px;
1848   background: #edf5f4;
1849   color: #467d78;
1850   padding: 4px 7px;
```

```
1851   border-radius: 3px;
1852   white-space: nowrap;
1853 }
1854 .subtle {
1855   color: var(--muted);
1856   font-size: 11px;
1857 }
1858 .instrument-panel > .panel-heading {
1859   background: #faffff;
1860   border-bottom: 1px solid var(--line);
1861 }
1862 .balance {
1863   width: 100%;
1864   height: auto;
1865   max-height: 330px;
1866   display: block;
1867   background-color: #f0f7fa;
1868   background-image:
1869     linear-gradient(#dfebf080 1px, transparent 1px),
1870     linear-gradient(90deg, #dfebf080 1px, transparent 1px);
1871   background-size: 24px 24px;
1872 }
1873 .pan-label {
1874   font-size: 12px;
1875   fill: #73909e;
1876   letter-spacing: 2px;
1877 }
1878 .beam {
1879   transition: transform 0.3s ease-out;
1880 }
1881 .instrument-readings {
1882   display: flex;
1883   border-top: 1px solid var(--line);
1884   padding: 15px 0;
1885   background: #fff;
1886 }
1887 .instrument-readings > div {
1888   flex: 1;
1889   display: flex;
1890   align-items: center;
1891   justify-content: center;
1892   gap: 15px;
1893   border-right: 1px solid var(--line);
1894 }
1895 .instrument-readings > div:last-child {
1896   border: 0;
1897 }
1898 .instrument-readings span {
1899   font-size: 11px;
1900   color: var(--muted);
```

```
1901 }
1902 .instrument-readings strong {
1903     font-size: 23px;
1904     font-family: Consolas, monospace;
1905     font-weight: 500;
1906     color: #234e63;
1907 }
1908 .instrument-readings small {
1909     font-size: 11px;
1910     margin-left: 5px;
1911     color: var(--muted);
1912 }
1913 .instrument-readings .status-word {
1914     font-family: inherit;
1915     font-size: 13px;
1916     color: var(--teal);
1917 }
1918 .feedback {
1919     display: flex;
1920     gap: 11px;
1921     background: #e7f3f2;
1922     border: 1px solid #d2e8e4;
1923     border-radius: 5px;
1924     padding: 12px 16px;
1925     margin: 14px 0;
1926     color: #40766e;
1927 }
1928 .feedback > span {
1929     font:
1930         600 11px Georgia,
1931         serif;
1932     width: 18px;
1933     height: 18px;
1934     border: 1px solid #75a79f;
1935     border-radius: 50%;
1936     text-align: center;
1937     line-height: 17px;
1938     flex-shrink: 0;
1939     margin-top: 1px;
1940 }
1941 .feedback strong {
1942     font-size: 11px;
1943 }
1944 .feedback p {
1945     font-size: 12px;
1946     margin: 3px 0 0;
1947 }
1948 .feedback.warning {
1949     background: #fff5e8;
1950     border-color: #efddc3;
```

```
1951   color: #916634;
1952 }
1953 .controls .panel-heading {
1954   padding-bottom: 12px;
1955 }
1956 .controls fieldset {
1957   border: 0;
1958   margin: 0;
1959   padding: 0;
1960   min-width: 0;
1961 }
1962 .control-grid {
1963   display: grid;
1964   grid-template-columns: 1fr 1fr 1.15fr;
1965   padding: 5px 20px 18px;
1966   gap: 18px;
1967 }
1968 .control-group {
1969   border-right: 1px solid var(--line);
1970   padding-right: 18px;
1971 }
1972 .control-group:last-child {
1973   border: 0;
1974   padding: 0;
1975 }
1976 .control-group h3 b {
1977   color: #8babb7;
1978   font:
1979     11px Consolas,
1980     monospace;
1981   margin-right: 4px;
1982 }
1983 .control-group > button.secondary,
1984 .done-button {
1985   padding: 8px 10px;
1986   width: 100%;
1987   font-size: 11px;
1988 }
1989 .done-button {
1990   background: #edf6f3;
1991   color: #367b6e;
1992   border: 1px solid #d7e9e1;
1993 }
1994 .control-group label {
1995   display: flex;
1996   align-items: center;
1997   justify-content: space-between;
1998   gap: 7px;
1999   font-size: 11px;
2000   color: #708691;
```

```
2001   margin: 8px 0;
2002 }
2003 .control-group select {
2004   font-size: 11px;
2005   padding: 5px 3px;
2006   max-width: 64%;
2007   width: 125px;
2008 }
2009 .slider-label {
2010   flex-wrap: wrap;
2011   margin-top: 13px !important;
2012 }
2013 .slider-label output {
2014   font-family: Consolas, monospace;
2015   color: #335c6c;
2016   font-size: 12px;
2017 }
2018 .control-group > small {
2019   font-size: 10px;
2020   color: #91a0a8;
2021   line-height: 1.5;
2022   display: block;
2023 }
2024 .weight-buttons {
2025   display: flex;
2026   gap: 6px;
2027 }
2028 .weight-buttons button {
2029   flex: 1;
2030   background: #f5f3eb;
2031   border: 1px solid #e5dfcb;
2032   border-radius: 5px;
2033   padding: 7px 3px;
2034   color: #7c6b49;
2035   font:
2036     14px Consolas,
2037     monospace;
2038 }
2039 .weight-buttons button span {
2040   display: block;
2041   font-size: 18px;
2042   line-height: 18px;
2043   color: #a28d5d;
2044 }
2045 .weight-buttons button small {
2046   font-size: 9px;
2047   margin-left: 2px;
2048 }
2049 .placed-weights {
2050   display: flex;
```

```
2051     gap: 4px;
2052     flex-wrap: wrap;
2053     min-height: 23px;
2054     padding-top: 6px;
2055 }
2056 .placed-weights button {
2057     font-size: 10px;
2058     background: #ecf2f5;
2059     color: #668492;
2060     padding: 3px 5px;
2061 }
2062 .placed-weights > span {
2063     font-size: 10px;
2064     color: #94a2aa;
2065 }
2066 .record-row {
2067     border-top: 1px solid var(--line);
2068     padding: 15px 20px;
2069     display: flex;
2070     align-items: center;
2071     gap: 10px;
2072     background: #fbfcfd;
2073 }
2074 .record-row label {
2075     font-size: 12px;
2076     color: #5d7988;
2077     margin-right: 4px;
2078     white-space: nowrap;
2079 }
2080 .unit-input {
2081     display: flex;
2082     border: 1px solid #d7e1e6;
2083     background: white;
2084     border-radius: 4px;
2085     align-items: center;
2086     width: 135px;
2087 }
2088 .unit-input input {
2089     border: 0;
2090     width: 100%;
2091     padding: 8px;
2092     font-family: Consolas, monospace;
2093 }
2094 .unit-input span {
2095     padding-right: 10px;
2096     font-size: 11px;
2097     color: #91a1a9;
2098 }
2099 .record-row button {
2100     font-size: 11px;
```

```
2101 padding: 9px 12px;
2102 white-space: nowrap;
2103 }
2104 .record-row > button:last-child {
2105   margin-left: auto;
2106 }
2107 .assessment {
2108   padding-bottom: 18px;
2109 }
2110 .assessment > .panel-heading {
2111   border-bottom: 1px solid var(--line);
2112 }
2113 .score {
2114   text-align: center;
2115   padding: 20px 0 4px;
2116 }
2117 .score strong {
2118   font:
2119     48px 'Bahnschrift',
2120     Consolas,
2121     monospace;
2122   color: var(--navy);
2123 }
2124 .score > span {
2125   font-size: 12px;
2126   color: #8ca0aa;
2127   margin-left: 5px;
2128 }
2129 .score p {
2130   font-size: 11px;
2131   color: #88a0aa;
2132   letter-spacing: 2px;
2133   margin: 3px;
2134 }
2135 .progress {
2136   height: 4px;
2137   background: #edf2f4;
2138   margin: 13px 23px 15px;
2139   overflow: hidden;
2140   border-radius: 4px;
2141 }
2142 .progress > div {
2143   height: 100%;
2144   background: var(--teal);
2145   transition: width 0.25s;
2146 }
2147 .step-list {
2148   list-style: none;
2149   margin: 0;
2150   padding: 0 21px;
```

```
2151 }
2152 .step-list li {
2153   display: flex;
2154   align-items: center;
2155   gap: 9px;
2156   border-bottom: 1px solid #eff3f5;
2157   padding: 13px 0;
2158   font-size: 11px;
2159   color: #637f8c;
2160 }
2161 .step-list li:last-child {
2162   border-bottom: 0;
2163 }
2164 .step-no {
2165   border: 1px solid #dce6ea;
2166   color: #96a9b2;
2167   font:
2168     10px Consolas,
2169     monospace;
2170   width: 24px;
2171   height: 24px;
2172   border-radius: 50%;
2173   display: grid;
2174   place-items: center;
2175   flex-shrink: 0;
2176 }
2177 .step-list li.passed .step-no {
2178   background: #e3f2ed;
2179   border-color: #e3f2ed;
2180   color: #1c8b75;
2181 }
2182 .step-list li.passed > span:nth-child(2) {
2183   color: #286c61;
2184 }
2185 .step-list li b {
2186   margin-left: auto;
2187   color: #4b7082;
2188   font:
2189     14px Consolas,
2190     monospace;
2191   white-space: nowrap;
2192 }
2193 .step-list li small {
2194   font-size: 10px;
2195   color: #9bafb8;
2196 }
2197 .assessment-note {
2198   padding: 0 21px;
2199   color: #94a5ad;
2200   font-size: 10px;
```

```
2201   margin: 13px 0 17px;
2202   line-height: 1.8;
2203 }
2204 .full {
2205   display: block;
2206   margin: 0 20px;
2207   width: calc(100% - 40px);
2208   font-size: 12px;
2209 }
2210 .event-panel {
2211   margin-top: 15px;
2212   padding: 0 19px 10px;
2213 }
2214 .event-panel .panel-heading {
2215   padding: 16px 0 8px;
2216 }
2217 .event {
2218   display: flex;
2219   gap: 10px;
2220   font-size: 10px;
2221   color: #7b8f99;
2222 }
2223 .event > i {
2224   width: 5px;
2225   height: 5px;
2226   border-radius: 50%;
2227   background: #88b1aa;
2228   flex-shrink: 0;
2229   margin-top: 14px;
2230 }
2231 .event > i.warn {
2232   background: #d0a26c;
2233 }
2234 .event p {
2235   margin: 7px 0;
2236   line-height: 1.6;
2237 }
2238 .event time {
2239   display: block;
2240   font-family: Consolas, monospace;
2241   font-size: 9px;
2242   color: #a3b0b7;
2243   margin-top: 3px;
2244 }
2245 footer {
2246   height: 45px;
2247   display: flex;
2248   align-items: center;
2249   justify-content: space-between;
2250   padding: 0 4%;
```

```
2251 font-size: 10px;
2252 color: #96a7b1;
2253 border-top: 1px solid #e3eae;
2254 background: #f7f9fa;
2255 }
2256 .start-panel {
2257   background: #fff;
2258   border: 1px solid var(--line);
2259   padding: 42px;
2260   max-width: 590px;
2261   margin: 35px auto;
2262   border-radius: 8px;
2263   display: flex;
2264   flex-direction: column;
2265   gap: 13px;
2266 }
2267 .start-panel h2,
2268 .start-panel p {
2269   margin: 0;
2270   text-align: center;
2271 }
2272 .start-panel p {
2273   font-size: 12px;
2274   color: var(--muted);
2275   margin-bottom: 13px;
2276 }
2277 .start-mark {
2278   text-align: center;
2279   font-size: 55px;
2280   color: #356979;
2281 }
2282 .start-panel label,
2283 .new-dialog label {
2284   display: flex;
2285   flex-direction: column;
2286   gap: 8px;
2287   color: #78909d;
2288   font-size: 12px;
2289 }
2290 .start-panel .primary {
2291   margin-top: 9px;
2292 }
2293 .history-panel {
2294   overflow: auto;
2295 }
2296 table {
2297   width: 100%;
2298   border-collapse: collapse;
2299   text-align: left;
2300   font-size: 12px;
```

```
2301 }
2302 th {
2303     background: #eef4f6;
2304     padding: 15px 20px;
2305     color: #658392;
2306     font-weight: 500;
2307     white-space: nowrap;
2308 }
2309 td {
2310     padding: 18px 20px;
2311     border-bottom: 1px solid #e8eef1;
2312 }
2313 td small {
2314     display: block;
2315     font:
2316         10px Consolas,
2317         monospace;
2318     color: #9aadb7;
2319     margin-top: 6px;
2320 }
2321 td strong {
2322     font-weight: 500;
2323 }
2324 .empty {
2325     text-align: center;
2326     padding: 75px 20px;
2327     color: #7a929e;
2328 }
2329 .guide-grid {
2330     display: grid;
2331     grid-template-columns: repeat(4, 1fr);
2332     gap: 18px;
2333 }
2334 .guide-grid section {
2335     padding: 25px;
2336 }
2337 .guide-no {
2338     font:
2339         36px 'Bahnschrift',
2340         Consolas,
2341         monospace;
2342     color: #a7c4cf;
2343 }
2344 .guide-grid h2 {
2345     font-size: 14px;
2346     display: flex;
2347     justify-content: space-between;
2348     align-items: center;
2349     margin-top: 20px;
2350 }
```

```
2351 .guide-grid h2 small {
2352     font-size: 10px;
2353     color: #789eaa;
2354     font-weight: 400;
2355 }
2356 .guide-grid p {
2357     font-size: 12px;
2358     color: #78909d;
2359 }
2360 .guide-disclaimer {
2361     font-size: 12px;
2362     color: #78909d;
2363     margin-top: 25px;
2364 }
2365 .error {
2366     background: #fff0e9;
2367     color: #9f5a37;
2368     padding: 12px;
2369     border: 1px solid #e7ccb7;
2370     margin-bottom: 20px;
2371     border-radius: 5px;
2372     font-size: 12px;
2373 }
2374 .error button {
2375     margin-left: 15px;
2376     color: #945632;
2377     background: white;
2378     padding: 5px;
2379 }
2380 .modal-backdrop {
2381     position: fixed;
2382     inset: 0;
2383     background: #15344466;
2384     backdrop-filter: blur(3px);
2385     display: flex;
2386     align-items: center;
2387     justify-content: center;
2388     z-index: 5;
2389     padding: 25px;
2390 }
2391 .new-dialog {
2392     background: white;
2393     padding: 30px;
2394     border-radius: 10px;
2395     width: 450px;
2396     max-width: 100%;
2397     box-shadow: 0 20px 80px #122b4133;
2398 }
2399 .new-dialog h2 {
2400     margin-top: 0;
```

```
2401 }
2402 .new-dialog p {
2403     font-size: 12px;
2404     color: #7a929e;
2405 }
2406 .new-dialog label {
2407     margin-top: 18px;
2408 }
2409 .dialog-actions {
2410     display: flex;
2411     justify-content: flex-end;
2412     gap: 9px;
2413     margin-top: 25px;
2414 }
2415 .report {
2416     background: white;
2417     width: 900px;
2418     max-height: 90vh;
2419     overflow: auto;
2420     padding: 36px;
2421     border-radius: 9px;
2422     box-shadow: 0 20px 80px #122b4133;
2423 }
2424 .report-tools {
2425     display: flex;
2426     justify-content: flex-end;
2427     gap: 9px;
2428     margin-bottom: 30px;
2429 }
2430 .report h1 {
2431     font-size: 24px;
2432 }
2433 .report > p {
2434     font-size: 12px;
2435     color: #8397a2;
2436 }
2437 .report-summary {
2438     background: #edf5f5;
2439     display: flex;
2440     padding: 23px;
2441     margin: 25px 0;
2442     gap: 45px;
2443 }
2444 .report-summary div {
2445     display: flex;
2446     flex-direction: column;
2447     gap: 10px;
2448 }
2449 .report-summary span {
2450     font-size: 11px;
```

```
2451   color: #78909d;
2452 }
2453 .report-summary strong {
2454   font:
2455     22px 'Bahnschrift',
2456     'Microsoft YaHei',
2457     sans-serif;
2458   color: #216563;
2459 }
2460 .report h2 {
2461   font-size: 15px;
2462   margin-top: 30px;
2463 }
2464 .report td,
2465 .report th {
2466   padding: 12px 14px;
2467   font-size: 11px;
2468 }
2469 .report-events td:first-child {
2470   white-space: nowrap;
2471   font-size: 10px;
2472 }
2473 .report-note {
2474   border-top: 1px solid var(--line);
2475   margin-top: 25px;
2476   padding-top: 15px;
2477 }
2478 @media (min-width: 1500px) {
2479   .balance {
2480     max-height: 380px;
2481   }
2482 }
2483 @media (max-width: 1150px) {
2484   .workbench {
2485     grid-template-columns: minmax(0, 1fr) 275px;
2486     gap: 15px;
2487   }
2488   .control-grid {
2489     gap: 10px;
2490     padding-left: 14px;
2491     padding-right: 14px;
2492   }
2493   .control-group {
2494     padding-right: 10px;
2495   }
2496   .panel-heading {
2497     padding: 15px;
2498   }
2499   .instrument-readings > div {
2500     gap: 8px;
```

```
2501 }
2502 .instrument-readings strong {
2503   font-size: 20px;
2504 }
2505 .record-row {
2506   padding: 13px;
2507   gap: 7px;
2508 }
2509 .record-row button {
2510   padding: 9px;
2511 }
2512 .unit-input {
2513   width: 95px;
2514 }
2515 .guide-grid {
2516   grid-template-columns: repeat(2, 1fr);
2517 }
2518 }
2519 @media (max-width: 850px) {
2520   .workbench {
2521     grid-template-columns: 1fr;
2522   }
2523   .workbench aside {
2524     display: grid;
2525     grid-template-columns: 1fr 1fr;
2526     gap: 15px;
2527   }
2528   .event-panel {
2529     margin-top: 0;
2530   }
2531   .connection {
2532     font-size: 0;
2533   }
2534   .navigation > span {
2535     display: none;
2536   }
2537   .brand strong {
2538     font-size: 14px;
2539   }
2540   .brand span {
2541     font-size: 9px;
2542   }
2543   .topbar {
2544     gap: 9px;
2545     padding: 0 20px;
2546   }
2547   .page-heading h1 {
2548     font-size: 23px;
2549   }
2550   .page-heading > button {
```

```
2551     position: static;
2552     margin-top: 15px;
2553   }
2554   main {
2555     padding: 22px;
2556   }
2557   .guide-grid {
2558     gap: 12px;
2559   }
2560 }
2561 @media (max-width: 550px) {
2562   .version {
2563     display: none;
2564   }
2565   .brand-icon {
2566     width: 34px;
2567     height: 34px;
2568     font-size: 22px;
2569   }
2570   .brand strong {
2571     font-size: 12px;
2572   }
2573   .brand span {
2574     letter-spacing: 1px;
2575   }
2576   .navigation nav {
2577     gap: 23px;
2578   }
2579   .workbench aside {
2580     display: block;
2581   }
2582   .event-panel {
2583     margin-top: 15px;
2584   }
2585   .control-grid {
2586     grid-template-columns: 1fr;
2587   }
2588   .control-group {
2589     border-right: 0;
2590     border-bottom: 1px solid var(--line);
2591     padding: 10px 0 18px;
2592   }
2593   .control-group label {
2594     font-size: 12px;
2595   }
2596   .record-row {
2597     flex-wrap: wrap;
2598   }
2599   .record-row > button:last-child {
2600     margin-left: 0;
```

```
2601 }
2602 .instrument-readings {
2603   flex-wrap: wrap;
2604 }
2605 .instrument-readings > div {
2606   flex-direction: column;
2607   gap: 5px;
2608 }
2609 .panel-heading > div {
2610   flex-wrap: wrap;
2611 }
2612 .guide-grid {
2613   grid-template-columns: 1fr;
2614 }
2615 .report {
2616   padding: 20px;
2617 }
2618 .report-tools {
2619   flex-wrap: wrap;
2620 }
2621 .report-summary {
2622   gap: 20px;
2623   padding: 15px;
2624 }
2625 .report-summary strong {
2626   font-size: 16px;
2627 }
2628 footer span:last-child {
2629   display: none;
2630 }
2631 .modal-backdrop {
2632   padding: 10px;
2633 }
2634 }
2635 @media (prefers-reduced-motion: reduce) {
2636   * {
2637     transition: none !important;
2638     scroll-behavior: auto !important;
2639   }
2640 }
2641 @media print {
2642   @page {
2643     size: A4;
2644     margin: 18mm;
2645   }
2646   :root,
2647   html,
2648   body,
2649   #root {
2650     background: #fff;
```

```
2651 }
2652 body:has(.report-backdrop) .topbar,
2653 body:has(.report-backdrop) .navigation,
2654 body:has(.report-backdrop) main,
2655 body:has(.report-backdrop) footer {
2656     display: none;
2657 }
2658 .report-backdrop {
2659     position: static;
2660     display: block;
2661     background: white;
2662     padding: 0;
2663     backdrop-filter: none;
2664 }
2665 .report {
2666     width: 100%;
2667     max-height: none;
2668     overflow: visible;
2669     box-shadow: none;
2670     border-radius: 0;
2671     padding: 0;
2672 }
2673 .report-tools {
2674     display: none;
2675 }
2676 .report-summary {
2677     print-color-adjust: exact;
2678 }
2679 .report h2 {
2680     break-after: avoid;
2681 }
2682 tr {
2683     break-inside: avoid;
2684 }
2685 .report h1 {
2686     font-size: 22px;
2687 }
2688 .report td,
2689 .report th {
2690     padding: 5px 8px;
2691     font-size: 10px;
2692 }
2693 .report p {
2694     margin-top: 8px;
2695     margin-bottom: 8px;
2696 }
2697 .report-events td:first-child {
2698     font-size: 9px;
2699 }
2700 thead {
```

```
2701     display: table-header-group;
2702   }
2703   .report-note {
2704     font-size: 10px;
2705   }
2706 }
2707 <!doctype html><html lang="zh-CN"><head><meta charset="UTF-8"/><meta name="viewport"
content="width=device-width,initial-scale=1.0"/><meta name="theme-color" content="#16374b"/><link
rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 32 32'%3E%3Crect
width='32' height='32' rx='6' fill='%2316374b'/%3E%3Ctext x='16' y='24' text-anchor='middle' fill='white'
font-size='25'%3E%3C/text%3E%3C/svg%3E"/><title>天平实验训练示例 V1.0</title></head><body><div
id="root"></div><script type="module" src="/src/main.tsx"></script></body></html>
2708 import { defineConfig } from 'vite';
2709 import react from '@vitejs/plugin-react';
2710 export default defineConfig({
2711   plugins: [react()],
2712   server: { proxy: { '/api': 'http://127.0.0.1:3187' } },
2713 });
2714 $ErrorActionPreference = 'Stop'
2715 $projectRoot = Split-Path -Parent $PSScriptRoot
2716 Set-Location -LiteralPath $projectRoot
2717 $nodeCommand = Get-Command node -ErrorAction SilentlyContinue
2718 if (-not $nodeCommand) { throw 'Node.js is missing. Install Node.js 22.18+ or 24 LTS, then retry.' }
2719 $nodeVersion = [version]((& $nodeCommand.Source --version).TrimStart('v'))
2720 if ($nodeVersion -lt [version]'22.18.0') { throw 'Node.js 22.18 or newer is required.' }
2721 if (-not (Test-Path -LiteralPath (Join-Path $projectRoot 'dist/index.html'))) { throw 'Missing
dist/index.html. Run npm ci and npm run build first.' }
2722 $port = 3187
2723 if ($env:PORT) { $port = [int]$env:PORT }
2724 if ($port -lt 1024 -or $port -gt 65535) { throw 'PORT must be 1024..65535.' }
2725 $url = "http://127.0.0.1:$port"
2726 $runtimeDir = Join-Path $projectRoot '.runtime'
2727 New-Item -ItemType Directory -Force -Path $runtimeDir | Out-Null
2728 $probe = New-Object System.Net.Sockets.TcpClient
2729 try { $probe.Connect('127.0.0.1', $port); $occupied = $true } catch { $occupied = $false } finally {
$probe.Dispose() }
2730 if ($occupied) { throw "Port $port is already in use. Open $url if this system is running, or stop the
occupying process. No process was stopped." }
2731 $serverPath = Join-Path $projectRoot 'server/index.ts'
2732 $process = Start-Process -FilePath $nodeCommand.Source -ArgumentList @('"' + $serverPath + '"')
-WorkingDirectory $projectRoot -WindowStyle Hidden -RedirectStandardOutput (Join-Path $runtimeDir
'server.log') -RedirectStandardError (Join-Path $runtimeDir 'server-error.log') -PassThru
2733 @{ pid = $process.Id; serverPath = $serverPath; startedAt =
$process.StartTime.ToUniversalTime().ToString('o'); port = $port } | ConvertTo-Json | Set-Content
-LiteralPath (Join-Path $runtimeDir 'server.json') -Encoding UTF8
2734 $ready = $false
2735 for ($attempt=0; $attempt -lt 40; $attempt++) {
2736   Start-Sleep -Milliseconds 250
2737   try { $health = Invoke-RestMethod -Uri "$url/api/health" -TimeoutSec 1; if ($health.status -eq 'ok') {
$ready = $true; break } } catch {}
2738   if ($process.HasExited) { break }
2739 }
2740 if (-not $ready) { throw "Startup failed. Read .runtime/server-error.log. PID: $($process.Id)" }
2741 Write-Host "Running: $url"
2742 if ($env:BALANCE_NO_BROWSER -ne '1') { Start-Process $url }
2743 $ErrorActionPreference = 'Stop'
2744 $projectRoot = Split-Path -Parent $PSScriptRoot
2745 $pidFile = Join-Path $projectRoot '.runtime/server.json'
2746 if (-not (Test-Path -LiteralPath $pidFile)) { Write-Host 'No recorded server process.'; exit 0 }
2747 $record = Get-Content -LiteralPath $pidFile -Raw -Encoding UTF8 | ConvertFrom-Json
2748 $process = Get-Process -Id $record.pid -ErrorAction SilentlyContinue
2749 if (-not $process) { Write-Host 'Server is already stopped.'; exit 0 }
2750 $command = Get-CimInstance Win32_Process -Filter "ProcessId = $($record.pid)"
2751 $recordStart = [datetime]$record.startedAt
2752 if ($process.ProcessName -ne 'node' -or -not $command.CommandLine.Contains($record.serverPath) -or
$process.StartTime.ToUniversalTime().Ticks -ne $recordStart.ToUniversalTime().Ticks) {
2753   throw 'Process identity changed. Refusing to stop a different process.'
2754 }
2755 Stop-Process -Id $record.pid
2756 Write-Host 'Server stopped. Experiment records are preserved.'
```